

Matter

1. Choose the best alternative.

1. Which of the following is negatively charged subatomic particles?

- (i) neutron (ii) electron (iii) proton (iv) none of them
- ✓ (ii) electron

2. Which of the following symbol represent to the element mercury?

- (i) Mo (ii) Mg (iii) Mn (iv) Hg
- ✓ (iv) Hg

3. Which of the following group in the modern periodic table represent to zero group?

- (i) IA (ii) IVA (iii) VIA (iv) VIIIA
- ✓ (iv) VIIIA

4. The members of group IA are called as

- (i) alkali metal (ii) alkali earth metal (iii) metalloids (iv) inert gases
- ✓ (i) alkali metal

5. Which of the following is the diatomic molecule?

- (i) Oxygen (ii) Sulphur (iii) Ozone (iv) Sodium
- ✓ (i) Oxygen

6. The number of protons (p^+), electrons (e^-) and neutrons (n^0) of various elements are given in the table below.

Element	No. of proton	No. of electron	No. of neutron
Potassium	19	19	20
Carbon	6	6	6
Oxygen	8	8	8
Argon	18	18	22

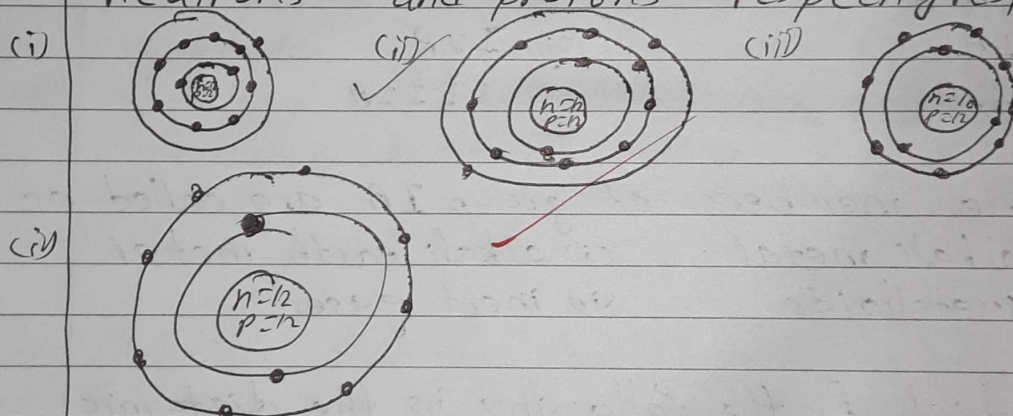
Which of these has the highest atomic mass?

- (i) Potassium (ii) Carbon
(iii) Oxygen (iv) Argon

7. Which of the following is true for an element?

- (i) atomic number = no. of protons + no. of electrons
(ii) atomic number = no. of protons + no. of neutrons
(iii) atomic number = no. of electron + no. of neutrons
(iv) atomic number = no. of ~~protons~~ protons + no. of electrons

8. Identify the Mg^{2+} ion from the figure where n and p represent the number of neutrons and protons respectively respectively



B) Very short answer questions

1) Define an element?

→ An element is a pure substance made up of only one kind of atom and cannot be broken up into two or more simple substances by any physical or chemical method.

2) Define Atomic number

→ The number of protons or electrons present in a neutral atom is called atomic number.

3) What is atomicity?

→ The number of atoms present in one molecule of an element is known as atomicity.

4) Define Radical.

→ A group of atoms (having some negative or positive charges on them) which can retain its identity in series of chemical reactions is called radical.

5) What is an atom?

→ An atom is the smallest particle of an element that can take part in a chemical reaction.

6) What is electronic configuration?

→ The systematic distribution of electrons in different shells of an atom is called electronic configuration.

7) What is valence electron?

→ The electron present in valence shell is called valence electron.

8) What is molecule?

→ The smallest particle of a substance (element or compound) that retains the properties of that substance and can exist in free state is called molecule.

9) What is molecular formula?

→ The molecule of a compound is represented by a symbol called molecular formula.

Sw
23/5/17

10) Write the molecular formula of the following compounds.

a) Sodium chloride $\rightarrow$ NaCl b) Aluminium chloride $\rightarrow$ AlCl₃

c) Calcium carbonate $\rightarrow$ CaCO₃ d) Magnesium nitride $\rightarrow$ ~~CaO~~ Mg₃N₂

e) Calcium oxide $\rightarrow$ CaO f) Nitric acid $\rightarrow$ HNO₃

g) Hydrochloric acid $\rightarrow$ HCl h) Sulphuric acid $\rightarrow$ H₂SO₄

i) Sodium hydroxide $\rightarrow$ NaOH j) Aluminium hydroxide $\rightarrow$ Al(OH)₃

11) Short answer questions.

1) Hydrogen is an element. Give reason:

$\rightarrow$ Hydrogen is an element because it is made up of only kind of atom and it cannot be broken down into simple substances.

2) Valency of carbon is 4. Give reason.

$\rightarrow$ Valency of carbon is 4 because there are total 8 electrons in ~~it~~ ^{409416/1051} there are 4 electrons in only in valence shell and it needs 4 more 4 electrons to complete its outer shell.

3) Write differences between atom and molecule.

→ Atom

(i) An atom is the smallest particle of an element.

(ii) It may exist alone or combine with other atoms.

(iii) Examples - H, O, C

Molecule

(i) A molecule is the smallest particle of a substance that can exist independently.

(ii) It is made up of two or more atoms joined together.

(iii) Exams - H_2 , O_2 , H_2O

4) Write the difference between $2H$ and H_2

→ $2H$

(i) It represents two separate hydrogen atoms.

(ii) It has two H atoms that are not joined.

H_2

(i) It represents one hydrogen molecule.

(ii) It has two H atoms joined together.

5) Water is compound. Give reason.

→ Because it is made up of hydrogen and oxygen chemically combined in a fixed ratio of 2:1.

6) Atom is neutral. Give reason.

→ An atom is neutral because it has an equal number of positively charged protons and negatively charged electrons. The charge of atom is '0'.

7) Write the ~~fundat~~ fundamental properties of subatomic particles.

particle	charge	Relative mass	Location
Proton	Positive (+1)	1	Nucleus
Neutron	No charge (0)	1	Nucleus
Electron	Negative (-1)	$1/1840$	Outside the nucleus

8) Write any two significances of valence electron.

- Valence electrons determine the valency of an element.
- Valence electrons take part in chemical reactions and bonding.

9) Write any two importance of molecular formula.

- It shows the types and number of atoms present in a molecule.
- It helps to calculate the molecular mass of substance easily.

10) Find the molecular weight of limestone, ammonia and water.

→ Limestone $\rightarrow \text{CaCO}_3$

$$\begin{aligned} &\rightarrow 1 \times \text{C} + 1 \times \text{P} + 3 \times \text{O} \\ &= 1 \times 40 + 1 \times 12 + 3 \times 16 \\ &= 40 + 12 + 48 \\ &= 100 \text{ amu} \end{aligned}$$

→ Ammonia = NH_3

$$\begin{aligned} &= 1 \times \text{at. wt. of N} + 3 \times \text{at. m. of H} \\ &= 1 \times 14 + 3 \times 1 \\ &= 14 + 3 \\ &= 17 \text{ amu} \end{aligned}$$

- water $\rightarrow$ H_2O
 - $= 2 \times \text{at. mass of H} + 1 \times O$
 - $= 2 \times 1 + 1 \times 16$
 - $= 2 + 16$
 - $= 18 \text{ amu}$

11) Atom numbers of P, Q, R and S are 1, 2, 7 and 11 respectively. Classify them as metals, non-metal and noble gases.

$\rightarrow$

- P(1) = Hydrogen $\rightarrow$ non-metal
- Q(2) = Helium $\rightarrow$ noble gas
- R(7) = Nitrogen $\rightarrow$ Non-metal
- S(11) = Sodium $\rightarrow$ metal

D) Long answer questions

1) Observe the given table and answer the following questions.

Elements	Electronic configuration
A	2, 6
B	2, 8, 1
C	2, 8, 7

a) Identify the elements A, B, and C?

$\rightarrow$

- A $\rightarrow$ Oxygen
- B $\rightarrow$ Sodium
- C $\rightarrow$ Chlorine

b) Identify metal and non-metal from elements A, B and C

$\rightarrow$

- A $\rightarrow$ Non-metal
- B $\rightarrow$ Metal
- C $\rightarrow$ Non-metal

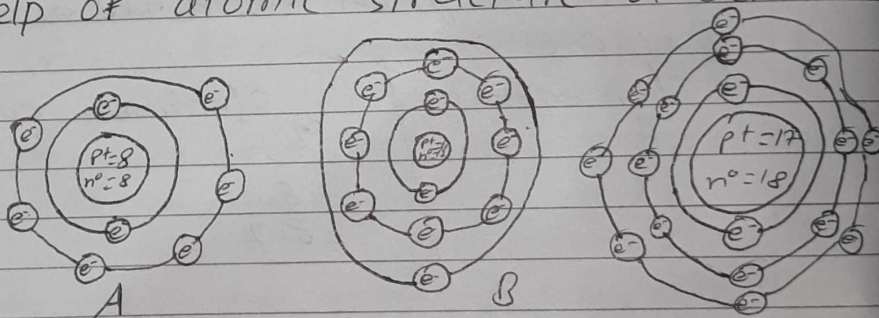
c) What is the valency of A, why?
→ The valency of 'O' is 2 because oxygen needs ~~2~~ 2 more electrons in its most shell.

d) Write the molecular formula of the compound formed by the combination of B and C.
→ NaCl .

e) Identify the group and periods of elements A, B, and C.

→ The group of element A is in Group 16, B is in Group 1 and C is in Group 18.
The period of element A is period 2, B is period 3 and C is period 3.

v) Answer the questions given below with the help of atomic structure of elements.



a) ~~Ident~~ Identify the elements A, B and C.

→ • A → Oxygen
• B → Sodium
• Argon C → Argon

b) Which group in the periodic table do the elements A, B and C occupy?

→ Group of A → Group 16
Group of B → Group 1
Group of C → Group 18

c) Write the valency of elements A, B and C.

→ The valency of element A is 2, B is 1 and C is 1.

d) What is the number of neutrons in elements A, B and C?

→ The number neutrons in elements A is 8, B is 12 and C is 18.

e) Observe the given table and answer the following questions.

Elements	Electronic configuration
A	2, 6
B	2, 8, 1
C	2, 8, 7
D	2, 8, 8, 1

a) Identify the elements A, B, C and D.

→

- A → Oxygen (O)
- B → Sodium (Na)
- C → ~~Argon (Ar)~~ Chlorine (Cl)
- D → Potassium (K)

b) Identify metal and non-metal from elements A, B, C and D.

→

- Metal → B, D
- Non-metal → A, C

c) What is the valency of A? Why?

→ The valency of A is 2 because its outer most shell needs two more electron to complete its valence shell. So, its valency is 2.

d) Write the molecular formula of the compound formed by the combination of C and D. Which bond is formed between them?

→ The molecular formula of the compound formed by the combination of C and D are KCl , and ionic bond is formed between them.

e) Which element is more active between B and D, why?

→ B is more active between B and D because ~~Na has one more electron shell~~
Na loses its outermost electron more easily.

4) Answer the following questions if atomic number of the element potassium is 19:

a) Write the electronic configuration of it on the basis of sub-shell.

→ $K = 1s^2 2s^2 2p^6 3s^2 3p^6 4s^1$
Total electron = $2 + 2 + 6 + 2 + 6 + 1 = 19$

b) Write its valence electron with reason.

→ Its valence electron is 1 because its valence shell have only one electron.

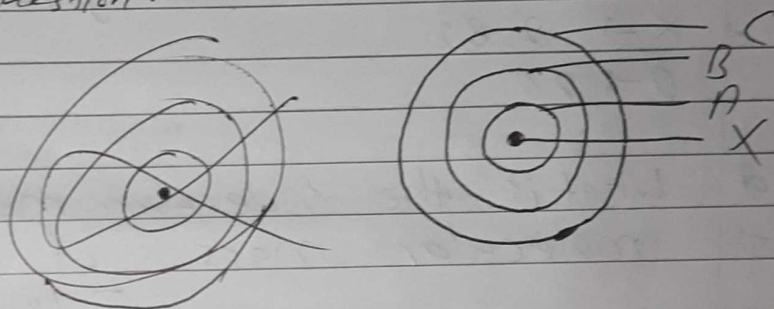
c) Draw the ionic structure of it.

→

d) It is a metal or non-metal? Write with reason.

→ It is a metal because it has 1 valence electron. It easily loses this electron to form K^+ ion and mainly it is shiny, soft, and a good conductor of heat and electricity.

5) The diagram given alongside represents structure of atom. study ~~the~~ it and answer the following question:



a) Name the shells denoted by A, B, C.

→ The name of shells denoted by A is K, B is L and C is M.

b) Which shell has minimum energy?

→ K shell has minimum energy.

c) Name X and state the charge on it.

→

d) Give the symbol of valence shell in the atom.

→

6) Two elements X and Y combine in a ratio of 3:2 by mass and the compound 'Z' is formed. Z is one of the essential components for human life. If Z is also one of the antacids, then

a) Identify X, Y and Z.

- • X → Magnesium (Mg)
- Y → Oxygen (O)
- Z → Magnesium oxide (MgO)

b) Write the electronic configuration of X and Y.

- X → 2, 8, 2
- Y → 2, 6

c) What is the molecular mass of Z?

- molecular mass of MgO.

$$= 1 \times \text{Mg} + 1 \times \text{O}$$

$$= (12 + 12) + (8 + 8)$$

$$= 24 + 16$$

$$= 40 \text{ amu}$$

d) Write the importance of Y in your life process.

- Importance of Y (Oxygen) in life process
 Oxygen is essential for respiration. It helps living organisms release energy from food for carrying out life processes.

* Unit * - GA. *

* The * End *